

MUHAMMAD IMAD KHAN

Data Science Undergraduate | Applied Machine Learning · Forecasting · Supply Chain Analytics
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EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Swabi, PK Sep 2023 – Present
Bachelor of Science in Data Science CGPA: 3.35/4.0

HONOURS

Finalist, UNCTAD Supply Chain Innovation Challenge 2026, UN Global Supply Chain Forum 2026
Best Intern among ~30 interns, US Mobile (2025); **Dean's Honour List**, GIKI

RESEARCH

Final Year Project: Cross-Firm Demand Forecasting for Collaborative Procurement 2026 – 2027
Official title: An Intelligent Multi-Enterprise Platform for Demand Forecasting, Collaborative Procurement and Supply-Chain Execution

- Tests whether pooled cross-firm order histories beat single-firm demand forecasts and whether pooling cuts procurement cost.
- Supervised by Dr. Shahabuddin Ansari; builds the matching and forecasting layers of Critical Mass with the walk-forward evaluation and SHAP explainability from AQI Predictor.

Critical Mass: A Digital Buying Cooperative for SMEs | *Technical Lead; UNCTAD finalist* 2026 – Present

- Formulated lot formation as a constrained matching problem over product, quantity, route, and delivery window, pooling SMEs' orders into container-sized lots.
- Designed sealed-bid supplier scoring on price, reliability, certification, proximity, and emissions, with split awards against supplier or route failure and Shapley-value allocation of savings.
- Sized the pilot from customs records (57,384 scrap shipments into Pakistan, \$1.25B, 752 importers); modelled 3-to-1 pooling to ~19,000 shipments, targeting 10–15% procurement savings.

EXPERIENCE

AI & Supply Chain Automation Intern, Qadri Group | [Prototype](#) | [Front End](#) Jun – Aug 2026

- Prototyped the platform on live PostgreSQL data (Streamlit, Plotly), coding supplier on-time delivery rates with sub-70% alerts, usage-based stock runway, and reorder levels from issuance rates and lead times.
- Built the React/TypeScript front end of QG-IRS, the team's production PostgreSQL/FastAPI platform, visualizing supplier spend Pareto, delivery delays, quoted versus actual freight, and dead stock over 90,000+ records.
- Audited every dashboard filter against live schemas, eliminating silent data mismatches.

AI Project Intern (Remote), 10Pearls | [Repository](#) | [Live Demo](#) Jun – Sep 2026

- Built a serverless 24/48/72-hour AQI forecaster for Lahore on 49,153 hourly weather and pollutant observations, automating ingestion, retraining, and serving with Hopsworks, GitHub Actions, and FastAPI.
- Engineered 33 features and benchmarked Ridge, Random Forest, Gradient Boosting, ARIMA, and MLP against persistence under purged walk-forward validation; the blend beat it at all horizons (R^2 0.83/0.75/0.72 vs 0.81/0.71/0.63).
- SHAP identified atmospheric pressure as the strongest driver after current AQI at 72 hours; documented methods and limitations in a technical report.

Data & Compliance Intern, US Mobile | Lahore, Pakistan (Hybrid) Jun – Sep 2025

- Screened 10M+ customer records (SQL, Python, Splunk, Elasticsearch) for anomalous, potentially fraudulent activity, and automated the ingestion and review workflow, saving 6+ hours weekly.

SELECTED PROJECTS

The Air We Breathe: Air Quality and Public Health | D3.js | [Repository](#) | [Live Demo](#) May 2026

- Scrollytelling analysis of 5,811 pollutant, weather, and health records; reports a null result ($r = -0.014$).

SKILLS

Analytics & ML: Python (Pandas, NumPy, scikit-learn, XGBoost, PyTorch), time-series forecasting, anomaly detection

Modelling & Evaluation: Feature engineering, walk-forward validation, model benchmarking, SHAP explainability

Data & Pipelines: SQL (PostgreSQL, Redshift), ETL, schema design, FastAPI, GitHub Actions (CI/CD), Hopsworks

Tools & Visualization: Git/GitHub, Linux/CLI, Docker, LLM API integration, Power BI, Tableau, D3.js, Plotly, Streamlit